

Incremental 3D Crop Model Association for Real-Time Counting in Dense Orchards

Daesung Park , KwangEun Ko , Dongbum Pyo , and Jaehyeon Kang 

Abstract—Accurate real-time crop counting is essential for autonomous agricultural systems. However, existing methods often fail in dense plantings due to heavy foliage, irregular planting patterns, and frequent occlusions. While 2D tracking suffers from double-counting and 3D reconstruction requires offline processing, we propose a real-time crop counting framework that incrementally constructs global 3D crop instances during data collection. Each crop is modeled as a 3D oriented bounding box, initialized upon detection and updated with subsequent observations. To ensure robust association across frames, we employ 3D Generalized Intersection over Union (GIoU) for spatial matching and confidence-based filtering for validation, effectively reducing double-counting in dense orchards. Unlike prior methods, our approach supports on-the-fly counting without post-hoc reconstruction and performs reliably in unstructured field conditions. Experimental results demonstrate the accuracy and real-time capability of the proposed system in dense agricultural settings.

Index Terms—Agricultural automation, robotics and automation in agriculture and forestry, object detection, segmentation and categorization.

I. INTRODUCTION

REAL-TIME crop counting is essential for autonomous agricultural systems, enabling responsive decision-making in tasks such as multi-robot coordination, targeted interventions, and adaptive harvesting. However, achieving precise counts in unstructured environments, such as natural orchards or open fields, remains challenging. Unlike controlled trellis or row-wise setups, unstructured settings lack regular plant arrangements and often feature dense foliage, high planting density, and minimal spacing between plants (e.g., 0.2 m). These factors lead to frequent occlusions and partial views of crops, significantly degrading counting performance.

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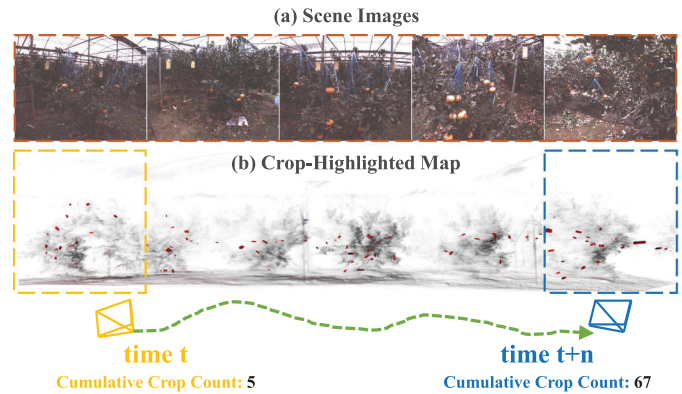


Fig. 1. Real-time incremental crop counting in densely planted environments. The proposed method enables real-time crop counting by combining 3D crop modeling with incremental data association within the 3D reconstruction process. Rather than relying on 2D tracking or post-hoc global reconstruction, the system continuously updates global 3D crop models on-the-fly. This approach offers a practical and real-time solution, addressing the limitations of 2D tracking (double-counting) and 3D reconstruction (offline processing and reconstruction failures) methods in dense and irregular cultivation settings.

Traditional 2D tracking-by-detection methods identify crops in image frames and associate them over time for automated crop counting. These frameworks support real-time processing due to their low computational cost. However, they are prone to double-counting due to tracking loss caused by occlusions or detection failures. This is particularly problematic in unstructured environments, where dense crops and irregular foliage cause frequent occlusions. Additionally, unpredictable scanning paths in such settings increase the likelihood of crop revisits, during which 2D trackers assign new instance IDs to previously observed crops, resulting in repeated counts.

To minimize double-counting, alternative approaches reconstruct a 3D model of the environment using techniques like simultaneous localization and mapping (SLAM) and structure-from-motion (SfM). However, these methods are unsuitable for real-time crop counting due to high computational demands from batch processing. Moreover, vision-based systems in unstructured agricultural settings suffer from frequent occlusions and weak visual features, leading to reconstructions with holes, noise, and misalignments. Additionally, the repetitive appearance of vegetation causes unreliable loop closure for SLAM. In particular, specialized sensors like RGB-D become ineffective in ambient sunlight, which severely degrades their depth-sensing capabilities. Although LiDAR is robust to these visual issues, it faces distinct challenges, as the constrained space in dense orchards amplifies noise from nearby objects, leading to accumulated mapping errors.

We propose a real-time, incremental 3D association framework using a LiDAR-Visual-Inertial configuration that avoids the pitfalls of both error-prone 2D tracking and offline 3D reconstruction (Fig. 1). Its parallelized architecture decouples state estimation from crop model processing, achieving the spatial perception of 3D methods while maintaining real-time performance (28 Hz). The system builds a persistent

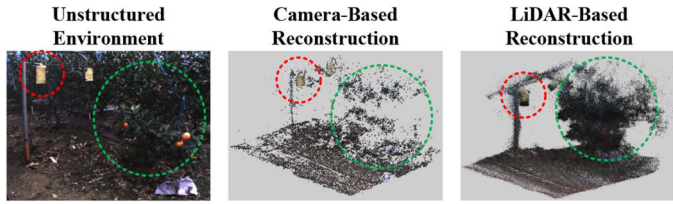


Fig. 2. Failure cases of 3D reconstruction in a densely planted environment with minimal space between plants. In the camera-based reconstruction (COLMAP [11]), the target plant (green circle) is incomplete due to limited image overlap and frequent occlusions. In the LiDAR-based reconstruction (FAST-LIVO [2]), noise increases in close-proximity plant regions, leading to accumulated errors that make the reconstructed plants and crops difficult to recognize.

map by modeling crops as 3D oriented bounding boxes. These models are initialized from filtered local scans and incrementally fused across frames by associating them via 3D Generalized Intersection over Union (GIoU), with confidence-based filtering applied to the fused detections. This incremental strategy eliminates double-counting, avoids batch processing delays, and ensures robustness against 3D reconstruction failures (Fig. 2), providing a practical real-time solution for crop counting.

The key contributions of our approach include:

- We introduce a crop counting method for densely planted environments with irregular arrangements, where existing 2D tracking and 3D reconstruction methods often fail.
- We represent each crop as a 3D bounding box, initialized at first detection using spatial filtering and geometric fitting, then updated with subsequent observations to handle viewpoint-dependent occlusions.
- We incrementally associate crops by spatially matching new detections to existing instances via 3D GIoU, then validate matches using confidence scores from the fused detections to ensure identity consistency over time.
- We design a real-time pipeline, which allows continuous crop count updates during scanning and highlights the practical value of the proposed system.
- We validate our framework's real-time counting accuracy in dense orchards through extensive experiments on a field dataset, which we make publicly available.

II. RELATED WORK

Image-based crop detection and tracking are widely used approaches for crop counting. Das et al. [3] employed SVM-based crop detection followed by optical-flow tracking to count crops in row-wise environments. Halstead et al. [4] performed IoU-based tracking, assuming one-directional robot path along rows in structured greenhouses with regularly spaced plants. Subsequent research [5] enhanced the tracking by incorporating wheel odometry-based reprojection and depth filtering for counting row-wise and trellised crops in glasshouses.

A comparative study by Villacr s et al. [6] evaluated multi-object tracking methods for crop counting in structured, row-wise apple orchards, finding that DeepSORT [7] achieved the lowest counting error. AgriSORT [8] extended appearance-based trackers like DeepSORT by integrating motion-based tracking with Kalman filtering for real-time multi-object tracking in structured, row-wise trellis vineyards. Similarly, CascadeSORT [9] addresses crop counting by combining Kalman filtering with cascade matching that leverages both appearance and motion features. Tu et al. [10] proposed a central region counting strategy to enhance the accuracy of tracking-based yield estimation by defining a robust region of interest.

These tracking-by-detection methods struggle with double-counting due to revisits, limiting their applicability to structured row-wise environments. This challenge is more significant in unstructured settings

with dense, irregular planting, where random scanning paths in uncontrolled environments lead to frequent crop revisits.

To address double-counting in tracking-based frameworks, several studies preprocess 3D map reconstruction before tracking. Liu et al. [11] combined crop pixel detection with object tracking using the Kanade-Lucas-Tomasi (KLT) tracker and Kalman filter, incorporating offline SfM-based 3D reconstruction to remove double-counting and outliers. Santos et al. [12] introduced a grape tracking method that leverages instance connectivity by projecting points from offline SfM onto 2D segmentation masks in row-wise trellis environments. In contrast, our approach enables real-time crop counting without relying on offline reconstruction.

Another approach involves reconstructing 3D maps before extracting crops from the reconstructed model. Liu et al. [13] used semantic SfM with crop and trunk pixels to convert 2D detections into 3D landmarks, preventing double-counting in row-wise orchards. Dong et al. [14] combined semantic segmentation and SfM map, identifying 3D crop points by intersecting segmented crop pixels with projected SfM points, followed by 3D connected component analysis [15] for apple counting in row-wise orchards. Fusaro et al. [16] used MinkPanoptic [17] to extract crop instances from colored point clouds in row-wise greenhouses. Meyer et al. [18] applied NeRF for multi-fruit counting using contrastive learning.

Unlike methods that extract crops from reconstructed maps, the proposed method achieves real-time crop counting by incrementally associating crop models within the 3D reconstruction process. Moreover, existing methods focus on controlled row-wise settings, whereas our approach targets unstructured environments with dense and irregular planting, where SfM and LiDAR-based reconstruction fail, as shown in Fig. 2.

Recent advances in 3D object detection (e.g., [19]), driven by autonomous driving, have shown promising results. However, the main challenge for applying these methods in agriculture is their need for 3D bounding box annotations—data that is common in autonomous driving but difficult to acquire in dense orchards. High levels of LiDAR noise from cluttered foliage make the 3D annotation process unreliable. Our framework avoids this bottleneck by operating with more accessible 2D annotations, providing a practical and deployable solution.

III. CROP COUNTING ALGORITHM

A. Overview

Fig. 3 illustrates the pipeline of the proposed real-time crop counting algorithm based on the incremental association of 3D crop models. Given camera, LiDAR, and IMU inputs, our method estimates sensor poses using a LiDAR-Inertial-Visual odometry algorithm (FAST-LIVO) [2]. It produces synchronized, posed image–point cloud pairs through tightly-coupled sensor fusion within an iterative Kalman filter framework.

The resulting posed images are used to estimate crop models in a global map. From each frame, crop instances with confidence scores are segmented using YOLOv11 [20]. The synchronized point cloud is then projected onto each segment to extract its instance-wise 3D points, and a 3D oriented bounding box is fitted to create the initial crop model. These per-frame models are incrementally associated with the global 3D crop map by measuring GIoU, either registered as new entities or merged with existing models in the crop model database. After confidence-based filtering, the remaining crop models in the global map determine the current count.

B. 3D Crop Model Generation

Each crop is represented by a 3D Oriented Bounding Box (3D OBB), providing a compact description of its position, orientation, and size. The OBB is parameterized by a center point, three orthogonal unit vectors defining its orientation (the box's local axes), and corresponding half-lengths along each axis. This representation supports robust

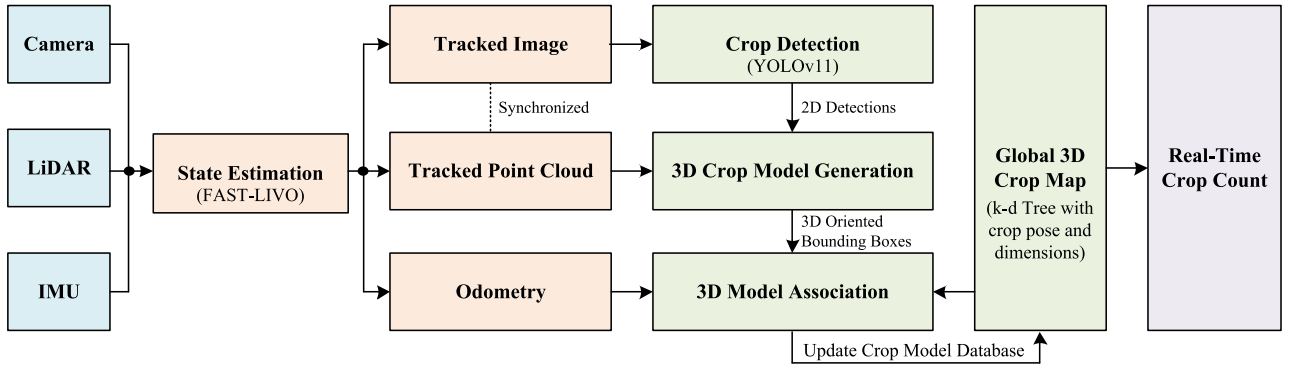


Fig. 3. Pipeline of the proposed real-time crop counting system. Camera, LiDAR, and IMU data are fused via FAST-LIVO [2] for state estimation. Tracked images and point clouds are used for crop detection and 3D crop model generation, with each crop represented as a 3D oriented bounding box. The generated models are incrementally associated with a global 3D crop map using GIoU, then filtered by segmentation confidence scores, enabling real-time crop counting in dense environments.

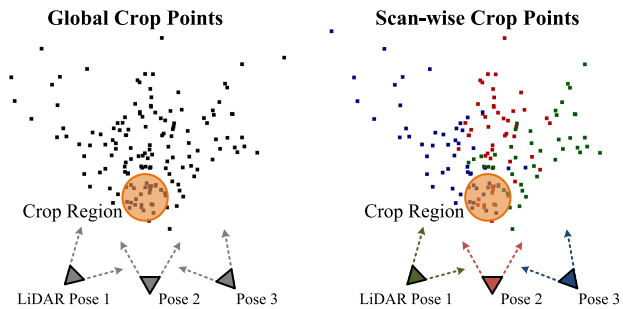


Fig. 4. Comparison of global and scan-wise crop points. Global crop points (left) exhibit accumulated noise due to high LiDAR noise near crops in densely planted orchards. In contrast, scan-wise crop points (right), colored by frame, capture localized observations from individual LiDAR poses, enabling more effective noise suppression.

association across viewpoints and effectively suppresses LiDAR noise, especially amplified in densely planted environments, through spatial filtering and geometric fitting.

Direct crop extraction from global point clouds suffers from noise accumulation, particularly in densely planted fields, where LiDAR noise is amplified by nearby vegetation and compounds over time. This leads to irregular noise artifacts around target crops, reducing the crop extraction accuracy from the global map. In contrast, constructing crop models from local per-frame scans systematically reduces such artifacts by keeping noise localized along the LiDAR ray direction. The proposed system filters each local scan before crop model generation, preventing cross-viewpoint noise accumulation. Fig. 4 illustrates a visual comparison of global and scan-wise crop points.

For local modeling, we obtain confidence-scored 2D instance segmentations using YOLOv11 [20]. We then extract 3D crop points by projecting the pre-calibrated LiDAR point cloud onto the image plane and selecting the points that fall within each segmentation mask. This process yields localized 3D point clusters corresponding to each crop instance.

We then apply a noise filtering step by removing points whose Euclidean distance to their nearest neighbor exceeds a predefined threshold (0.1 m in this work). To ensure reliable bounding box estimation, only clusters with at least five points remaining after filtering are used. For each valid cluster, the crop model center is defined as the centroid of the filtered points. The 3D OBB orientation is estimated using Principal Component Analysis (PCA), implemented via the Point Cloud Library (PCL) [21], with the principal axes forming the local coordinate frame of the OBB. The bounding box dimensions are then determined by

computing the minimum and maximum extents of the filtered points along these axes.

Each generated 3D OBB encapsulates a unique crop ID, position, orientation, dimensions, and segmentation confidence score. These bounding boxes are incrementally integrated and updated as new observations are acquired, enabling real-time crop tracking and counting. To support crop model updates, the point cloud associated with each crop instance is preserved and cumulatively merged with subsequent observations from different viewpoints. This fusion enhances model completeness and strengthens association reliability. The criteria for identifying whether two observations correspond to the same crop instance are detailed in the following section.

C. Incremental Association

The proposed method incrementally associates newly generated 3D crop models with existing models in the global crop map to enable accurate crop counting. To achieve this, each local 3D crop model, derived from each consecutive image–point cloud pair, is first transformed into the global coordinate frame using the corresponding odometry pose estimated by FAST-LIVO [2]. The transformed crop models are then evaluated for association with the existing global models, thereby preventing redundant crop representations and double-counting.

The association process consists of two main stages: candidate search and association decision. During candidate search, the algorithm identifies the nearest existing crop model C_{ref} in the global map for each newly generated crop model C_{new} by calculating the Euclidean distance between their centroids. To support an efficient nearest-neighbor retrieval, we employ a k-d tree to index and manage the global crop model database. This localized search strategy ensures scalability and real-time performance as the global map grows.

In the association stage, we determine if a new crop model (C_{new}) matches an existing model (C_{ref}) by calculating their 3D GIoU, a metric that evaluates spatial alignment across translation, rotation, and scale. If the GIoU exceeds a specified threshold, the models are associated: their point clouds are merged, a new 3D OBB is fitted to the combined points, and confidence scores are concatenated. The resulting fused model replaces C_{ref} in the global k-d tree. If no valid match is found, the newly generated crop model C_{new} is inserted into the global map as a new entry.

Unlike traditional IoU, which is limited to axis-aligned boxes, 3D GIoU accurately evaluates arbitrarily oriented 3D bounding boxes using convex hull-based computations, making it ideal for our 3D tracking task. The robustness of this metric allows our system to reliably associate crop detections across consecutive frames, thereby compensating for false negatives that may occur in any single frame.

To address false positive detections, we filter fused crop models based on their accumulated confidence scores from 2D segmentations.

Algorithm 1: AssociateCropModel.

Input: A new crop model C_{new} , a global crop model database indexed by $KDTree$, minimum number of observed frames N , confidence threshold τ_{conf} , and GloU threshold τ_{GloU} .
Output: Updated $KDTree$ with associated crop models.

```

 $C_{ref} = KDTree.NearestNeighbor(C_{new})$ 
if  $GloU(C_{new}, C_{ref}) > \tau_{GloU}$ 
  // Merge and update if associated
   $C_{fuse} \leftarrow Initialize()$ 
   $C_{fuse}.pcd = \{C_{ref}.pcd, C_{new}.pcd\}$ 
   $C_{fuse}.conf = \{C_{ref}.conf, C_{new}.conf\}$ 
   $C_{fuse}.obb = FitOrientedBoxUsingPCA(C_{fuse}.pcd)$ 
   $KDTree.Remove(C_{ref})$ 
   $KDTree.Insert(C_{fuse})$ 
if  $|C_{fuse}.conf| > N$  and  $median(C_{fuse}.conf) < \tau_{conf}$ 
   $KDTree.Remove(C_{fuse})$ 
end if
else
  // Add if not associated
   $KDTree.Insert(C_{new})$ 
end if
return  $KDTree$ 

```

A model is only considered for removal after it has been tracked for a minimum number of $N = 9$ frames to gather sufficient evidence. It is then filtered out if the median of its accumulated confidence scores falls below a predefined threshold. This filtering strategy effectively removes unreliable models and improves counting accuracy over time.

Crop counting is achieved by determining the number of crop models in the global map. The position and orientation of each crop can also be directly retrieved from its associated 3D OBB. The complete association procedure is summarized in Algorithm 1, which details candidate selection, decision logic, and k-d tree updates used to manage the global crop map.

D. Real-Time System Implementation

A key advantage of the proposed incremental crop counting framework is its ability to perform on-the-fly estimation as sensors scan the environment. Achieving this requires the entire pipeline to operate in real-time. To meet this requirement, the system adopts a lightweight, parallelized architecture that decouples crop model processing from state estimation tasks.

The system consists of two primary threads that operate in parallel. The state estimation thread, executed on a multi-core CPU, estimates poses for synchronized images and point clouds using FAST-LIVO [2]. These outputs are passed to the crop counting thread for further processing.

The crop counting thread handles all crop model-related operations, including crop detection, model generation, incremental association, and filtering. It receives tracked images and performs crop instance segmentation using YOLOv11, which is converted to ONNX format for efficient GPU inference. This conversion significantly reduces memory consumption and improves inference speed, enabling real-time processing of incoming frames without compromising accuracy. Additionally, a k-d tree structure is employed to efficiently manage and query the global crop model database during association.

Leveraging this architecture, the system performs real-time crop counting by processing local scans and incrementally updating the global crop map. This approach offers a real-time alternative to methods that rely on full global reconstruction or batch processing, which are computationally intensive and unsuitable for real-time deployment.

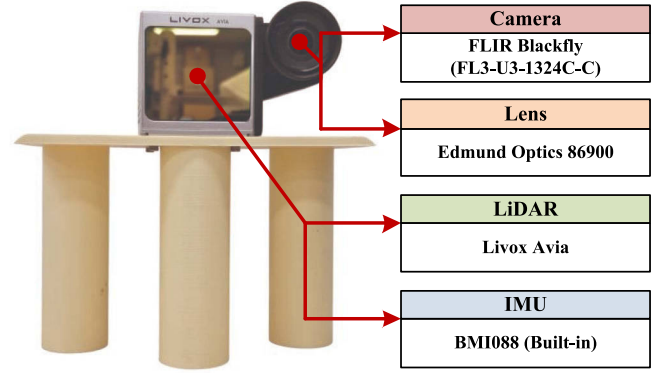


Fig. 5. Handheld sensor system used for data collection in densely planted environments. The platform consists of a FLIR Blackfly RGB camera (FL3-U3-1324C-C) with a 4.5 mm fixed focal length lens (Edmund Optics 86900) and a Livox Avia LiDAR with an integrated BMI088 IMU. This configuration captures high-resolution images, LiDAR point clouds, and inertial measurements for crop model construction and association.

IV. EXPERIMENTAL EVALUATION

A. Dataset

We used a custom handheld rig equipped with sensors to capture data in densely planted, unstructured environments, as shown in Fig. 5. The platform consisted of a FLIR Blackfly RGB camera with an Edmund Optics 86900 lens (providing 1280×1024 resolution, 84.7° horizontal FoV) and a Livox Avia LiDAR with an integrated BMI088 IMU. To ensure accurate cross-modal alignment, extrinsic calibration between the camera and LiDAR was performed using an edge-based method [22] that extracts and aligns edge features from both image and point cloud domains.

To evaluate our method under challenging and realistic conditions, we collected a dataset using the described sensor system. The data was captured in tangerine orchards (specifically Citrus unshiu and Kanpei) located in Seogwipo, Jeju, Republic of Korea. These orchards featured mature trees with dense foliage, and the target crops were in a pre-harvest state, showing a distinctive orange color and ranging from 5 to 10 cm in diameter and highly variable sensor-to-crop distances (0.2 to 3.0 m).

In these settings, we constructed a dataset¹ to evaluate the proposed crop counting system under challenging, real-world conditions. The dataset includes three representative motion patterns: (i) Revisit Sequences, where the sensors revisit a previously traversed region; (ii) Circular Sequences, representing rotational movement around a tree in confined spaces; and (iii) Straight Sequences, involving linear motion toward trees. A schematic diagram visualizing these trajectories is provided with the public dataset. For ground truth, frame-wise crop counts were manually annotated for each sequence, with details summarized in Table I. Fig. 6 illustrates the challenging conditions of the dataset, including uncontrolled illumination, dense foliage, and irregular vegetative patterns.

B. Evaluation

To evaluate the performance of the proposed method, we compare it against several representative baselines. For 2D tracking-based approaches, we include ByteTrack [23] and DeepSORT [7], two widely adopted multi-object tracking (MOT) algorithms. DeepSORT, in particular, has demonstrated strong performance in orchard environments, achieving the lowest error in a recent apple tracking study [6]. To benchmark against methods tailored explicitly for crop tracking, we

¹The dataset is accessible at <https://doi.org/10.5281/zenodo.15893214>.

TABLE I
DATASET CHARACTERISTICS

Sequence	Environment	Crop	Total Frames	Crop Count	Acquisition Date	Illumination*
Revisit 1	Open Field	Citrus unshiu	665	108	2024-12-19 AM	Cloudy
Revisit 2	Open Field	Citrus unshiu	607	110	2024-12-19 AM	Cloudy
Revisit 3	Green House	Kanpei	717	136	2024-12-20 PM	Neutral
Circular 1	Green House	Kanpei	500	37	2024-01-31 PM	Neutral
Circular 2	Green House	Kanpei	815	57	2024-02-01 AM	Neutral
Straight 1	Green House	Kanpei	611	69	2024-01-31 PM	Neutral
Straight 2	Green House	Kanpei	1389	92	2024-02-01 AM	Neutral
Straight 3	Open Field	Citrus unshiu	632	116	2024-12-20 AM	Sunny

* In the Green House environment, 'Neutral' illumination refers to the diffused natural light through the vinyl sheeting.

TABLE II
CROP COUNTING RESULTS OF OBJECT TRACKING, CROP TRACKING, AND MAP-BASED METHODS. BOLD INDICATES THE BEST PERFORMANCE.

Sequence	Object Tracking		Crop Tracking		Map-based				GT
	ByteTrack [23]	DeepSORT [7]	CascadeSORT [9]	AgriSORT [8]	CCA [14]	Distance	w/o Conf.	Proposed	
Revisit 1	2399	5603	1623	1733	75	133	123	105	108
Revisit 2	2431	5079	1594	2304	75	143	125	104	110
Revisit 3	3404	6128	1745	2321	81	159	144	122	136
Circular 1	385	898	435	355	30	69	35	34	37
Circular 2	626	1450	589	652	45	85	67	55	57
Straight 1	643	1031	599	336	47	87	85	67	69
Straight 2	1896	3665	1314	772	71	120	109	88	92
Straight 3	1491	2609	855	874	78	154	139	97	116
RMSE	1845.1	3764.6	1109.5	1313.0	31.4	28.7	14.5	8.9	-
Error Std. Dev.	1038.3	2089.9	505.7	801.7	15.5	6.3	7.5	6.4	-

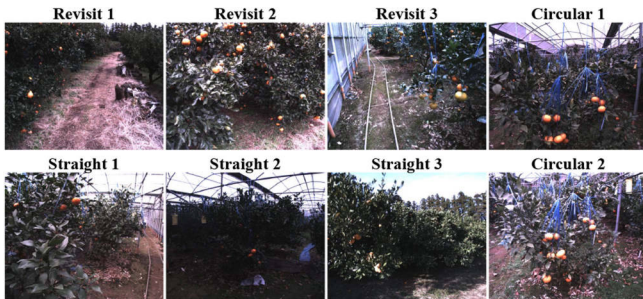


Fig. 6. Example images of the benchmark dataset. Data were collected from tangerine orchards in greenhouse and open-field settings. These unstructured environments feature densely planted trees and irregular planting patterns, leading to frequent occlusions and limited sensor visibility.

also include AgriSORT [8] and CascadeSORT [9]. All methods are evaluated using their publicly available implementations with default parameter settings.

To ensure a fair comparison that isolates the association strategy, we used a single YOLOv11s-seg detector for all baselines and the proposed method. We trained this model, initialized with COCO pre-trained weights, on 1,449 custom images (70% train, 10% val, 20% test) with 13,587 annotated crop masks, kept separate from our benchmark sequences. The model was trained for 100 epochs with a batch size of 32 using default hyperparameters from the official repository, and the final checkpoint was selected based on YOLOv11's validation metric ($0.1 \times \text{mAP}@0.5 + 0.9 \times \text{mAP}@0.5:0.95$).

As a reproducible 3D reconstruction baseline, we selected the Connected Component Analysis (CCA) method from [14]. To ensure a fair comparison, we applied CCA to the 3D map generated by our state estimator (FAST-LIVO [2]). Since the original work does not specify CCA parameters, we configured CCA with an octree level of 9 and a minimum cluster size of 5 points, aligning with our 3D OBB

generation requirements. This high-resolution setting preserves small object structures essential for accurate crop-level segmentation.

To further evaluate our association strategy, we implement two ablation baselines: one replacing GIoU with Euclidean distance between bounding box centroids, and another using GIoU without confidence-based filtering. These baselines isolate the individual contributions of spatial matching and confidence filtering to identity consistency. The association thresholds (τ_{conf} and τ_{GIoU} in Algorithm 1) were tuned using a dedicated validation sequence that is excluded from the benchmark test set, with final values set to $\tau_{conf} = 0.6$ and $\tau_{GIoU} = 0.25$. These thresholds were set constant across all benchmark sequences. While per-sequence tuning may yield marginal gains, using a single fixed parameter set highlights the generalizability of our method.

Table II summarizes the crop counting results of the proposed method compared to the baseline approaches. Overall, the proposed method achieves crop count estimates closest to the ground truth (GT), with the lowest RMSE of 8.9 and an error standard deviation of 6.4. These results demonstrate the accuracy and reliability of the proposed method in densely planted, unstructured agricultural environments.

Object tracking methods such as ByteTrack [23] and DeepSORT [7] exhibit significantly higher errors, primarily due to their reliance on 2D image-domain information. This reliance becomes a critical limitation in unstructured settings, where dense foliage, high planting density, frequent occlusions, and irregular sensor motion adversely affect appearance-based tracking performance. Although crop-specific trackers such as CascadeSORT [9] and AgriSORT [8] outperform general-purpose object trackers, their accuracy remains unsatisfactory, implying continued vulnerability to visual ambiguity and unstable associations in densely planted scenes.

Among the map-based methods, CCA [14] exhibits the highest error. This is primarily due to the accumulated pose drift and reconstruction errors in unstructured environments, where uneven terrain, dense vegetation, and irregular sensor motion introduce substantial challenges. These factors lead to noisy and distorted crop representations (as introduced in Fig. 2) that obscure crop boundaries and disrupt reliable

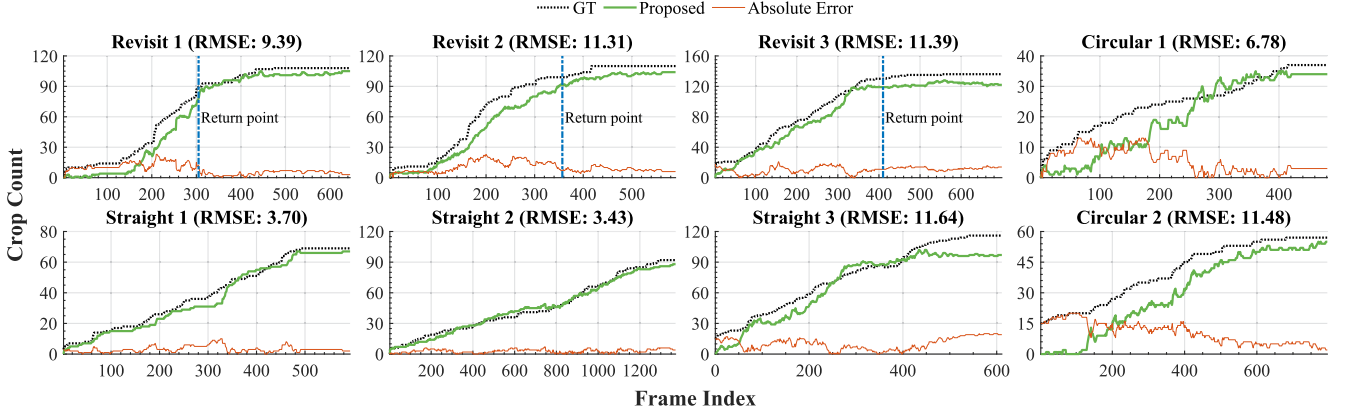


Fig. 7. Frame-wise crop count for each evaluation sequence. The dashed black line represents the ground truth (GT), the solid green line shows the cumulative crop count estimated by the proposed method, and the red line indicates the absolute frame-wise error. In the Revisit sequences, the vertical blue line marks the return point where the sensors begin revisiting the previously scanned area. Each subplot includes the RMSE and visualizes the cumulative crop count progression. While the proposed method closely follows the overall GT trend, counting delays are observed due to sparse LiDAR point density at longer ranges, particularly for small or distant crops.

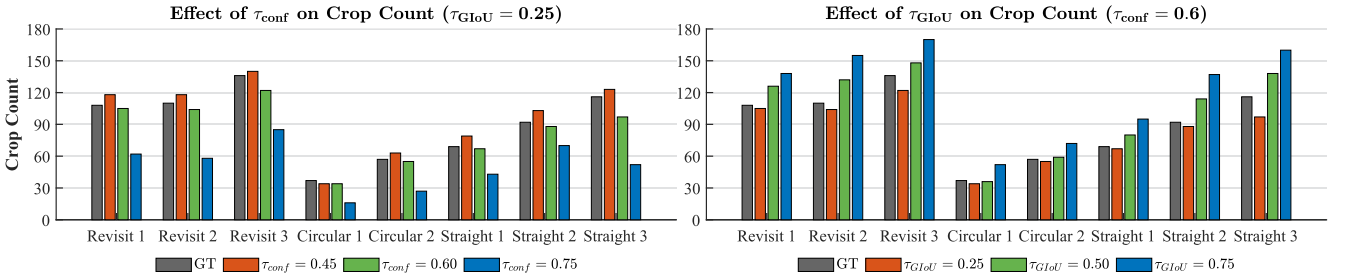


Fig. 8. Crop counting results across various threshold values for τ_{conf} (left) and τ_{GIoU} (right). Each bar group represents the estimated crop count for a given sequence and threshold setting, with the ground truth (GT) included for reference. In the left plot τ_{conf} is varied while τ_{GIoU} is fixed at 0.25; in the right plot, τ_{GIoU} is varied with τ_{conf} fixed at 0.6.

segmentation. Consequently, the effectiveness of the map-based approach is reduced in densely planted settings. In contrast, the proposed incremental approach overcomes these challenges by estimating crop models from localized scans, which preserve local structure and enable more accurate segmentation.

In the ablation studies, the distance baseline tends to overcount crops due to localization noise, which causes repeated observations of the same crop to fall outside the distance threshold (0.03 m). Similarly, removing confidence-based filtering degrades performance, with RMSE rising from 8.9 to 14.5 and standard deviation from 6.4 to 7.5. These results demonstrate that both GIoU-based association and confidence filtering are essential for accurate crop model association, as discussed in Section III-C.

C. Frame-Wise Crop Counting Analysis

To demonstrate the on-the-fly crop counting capability of the proposed method, we evaluate frame-wise crop counts across each evaluation sequence. Ground truth (GT) counts are obtained by manually annotating newly visible crops as they become observable in the image sequence. Fig. 7 shows the estimated crop counts over time, along with the corresponding ground truth and absolute error for each sequence.

Overall, the proposed method effectively tracks the general trend of the GT crop count. However, a noticeable delay in counting is observed compared to the GT annotations. This delay is primarily attributable to the modality gap between vision and LiDAR: crops farther from the sensor may appear in RGB images (and are thus included in the GT) but lack sufficient LiDAR points for reliable 3D crop modeling. This arises from the inherently low point density of LiDAR at longer ranges, particularly for small objects like crops. Consequently, these crops are

typically counted only when the platform moves closer or observes them across multiple frames, at which point they can be reliably modeled using both modalities.

Additionally, in the Revisit sequences, both the GT and estimated crop counts continue to increase gradually even after the initial traversal of the scene. This behavior is attributed to previously occluded crops or those outside the sensor's field of view becoming visible along the return path.

D. Effect of Association Thresholds on Crop Counting

As summarized in Algorithm 1, the proposed crop model association relies on two key parameters: a threshold on the confidence score (τ_{conf}), and a threshold on the 3D GIoU (τ_{GIoU}) between the two models. To evaluate the influence of these parameters on system performance, Fig. 8 presents the crop counting results under various threshold settings. The left plot varies τ_{conf} while fixing $\tau_{GIoU} = 0.25$, and the right plot varies τ_{GIoU} with $\tau_{conf} = 0.6$ fixed.

In the left plot, increasing τ_{conf} results in a consistent decline in crop count estimates across all sequences. This trend arises because a larger confidence score threshold filters out more potential crops, including valid detections. Consequently, the system tends to undercount when τ_{conf} is too high (e.g., 0.75), especially in dense scenes where valid detections may have lower confidence scores.

Conversely, the right plot shows that higher values of τ_{GIoU} lead to a steady increase in crop count estimates. A larger GIoU threshold makes the system more conservative, requiring more substantial spatial overlap to associate models. While this reduces false merges, it also increases the risk of under-association, where repeated observations of the same crop are not merged, resulting in overcounting.

TABLE III
AVERAGE EXECUTION TIME PER FRAME

	Time (ms)
Object Detection (YOLOv11)	16.09
Crop Model Generation (3D OBB)	0.41
Crop Model Association	0.63
Crop Map Update	18.17
Total Time	35.30

These results highlight a clear trade-off where a high τ_{conf} leads to undercounting, while a high τ_{GIoU} causes overcounting. Our experiments show that moderate thresholds ($\tau_{conf} = 0.6$ and $\tau_{GIoU} = 0.25$) achieve a desirable trade-off that minimizes total counting error across diverse sequences.

E. Runtime Analysis

To analyze the runtime of the proposed method, we report the average execution time of each module within the crop counting system. All experiments were conducted on a machine equipped with an 8-core Intel i7-11800H CPU, 16 GB RAM, and an NVIDIA GeForce RTX 3050Ti GPU, using the evaluation dataset described in Section IV-A.

The average runtime for each module is summarized in Table III, where each value represents the time required to process an image frame and its corresponding point cloud. Notably, the object detection runs exclusively on the GPU, while the remaining modules—including crop model generation, association, and map updating—are processed on the CPU. The total processing time per frame is 35.30 ms on average, corresponding to approximately 28.3 frames per second, satisfying real-time performance. These results demonstrate that the proposed system is both efficient and practical for deployment in real-world agricultural environments.

F. Limitations

While the proposed method demonstrates robust performance in densely planted, unstructured environments under typical conditions, it has several limitations. First, the system's reliance on RGB image-based detection makes it vulnerable to adverse weather and lighting conditions; performance may degrade in scenarios such as rain, nighttime operation, or intense sunlight, causing lens glare. Second, the framework focuses on incremental crop model association and does not incorporate global loop closure within the state estimation pipeline, which may affect long-term consistency in large-scale deployments. Third, although the crop models provide reliable geometric features, they are not leveraged as landmarks for odometry. Incorporating them into the state estimation—along with their spatial relationships—could improve localization accuracy. Lastly, occlusion due to camera–LiDAR viewpoint disparity is not explicitly modeled.

V. CONCLUSION

This paper presents a real-time crop counting method in densely planted environments, where conventional 2D tracking and global 3D reconstruction approaches often struggle due to frequent occlusions, irregular crop arrangements, and limited sensor visibility. The proposed method constructs robust 3D crop models by combining locally filtered LiDAR scans with 2D image segmentations to generate 3D Oriented Bounding Boxes. An incremental association strategy using 3D GIoU for spatial matching and confidence-based filtering for validation is introduced to maintain crop identity across frames while avoiding double-counting. This approach enables reliable crop tracking and counting in real-time. Experimental results on a field-collected dataset demonstrate that the proposed method achieves accurate and efficient crop counting under challenging agricultural conditions.

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